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FACULTY OF SCIENCE & TECHNOLOGY

DEPT. OF COMPUTER SCIENCE & ENGINEERING (CSE)

Project Proposal

Title:

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Submitted To:

Iyolita Islam & Rumana Yasmin

Lecturer of Department of CSE

Submitted By:
Group No: 3

Section: A

Name: Abdullah Al Galib & ID: 2252421049

Name:Md. Refayatul Islam & ID: 2252421059

Name: Paromita Chanda & ID: 2252421089

Name: Meghla Akter & ID: 2252421103

Abstract

Dhaka's air quality monitoring relies on a sparse sensor network, providing city-wide averages that fail to capture neighborhood-specific pollution spikes. To address this, we propose DeepAir-Dhaka, a spatiotemporal deep learning framework. Our approach utilizes Inverse Distance Weighting (IDW) to generate high-resolution virtual grids from limited ground data and employs a ConvLSTM network to predict future PM2.5 diffusion patterns. Unlike traditional models that output a single scalar value, DeepAir-Dhaka forecasts dynamic pollution heatmaps, enabling precise identification of localized hotspots and supporting targeted mitigation strategies in resource-constrained urban environments.

Keywords: Air Quality, ConvLSTM, Spatiotemporal Forecasting, Deep Learning, Dhaka.

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1. Introduction

Air pollution is one of the most critical challenges facing Dhaka city, consistently ranking it among the most polluted cities globally. While the Air Quality Index (AQI) provides a general idea of the city's pollution levels, it fails to capture the localized variations across different neighborhoods. For instance, the pollution level in a high-traffic area like Farmgate can be significantly higher than in a greener area like Ramna Park at the same time.

Current prediction models mostly focus on "Temporal Forecasting"—predicting a single AQI number for the entire city based on historical data. However, for effective urban planning and public health protection, this is insufficient. Citizens and policymakers need to know *where* the pollution is spreading, not just *when* it will rise.

This project proposes **"DeepAir-Dhaka,"** a Machine Learning system that moves beyond single-value predictions. By combining ground station data with grid-based mathematical interpolation, we aim to visualize pollution as a dynamic "heatmap." We will employ Deep Learning techniques to forecast how these pollution hotspots move across the city (Spatiotemporal Forecasting), providing a granular, neighborhood-level view of air quality.

2. Literature Review

Research on air quality prediction in Dhaka has primarily focused on traditional regression and time-series models.

- **Traditional Approaches:** Studies have extensively used Linear Regression, Random Forest, and standard LSTM (Long Short-Term Memory) networks to predict PM2.5 levels. These studies typically rely on data from the sparse network of CAMS (Continuous Air Monitoring Stations) or the US Consulate sensor.
- **Limitations:** The major gap in existing literature is the lack of spatial resolution. Most models treat Dhaka as a single point, ignoring the complex diffusion of pollutants caused by wind, traffic patterns, and building density.
- **Our Approach:** Recent global research suggests that combining interpolation techniques (like Inverse Distance Weighting) with Convolutional Neural Networks (CNNs) can effectively map pollution in areas without sensors. Our project adopts this "Spatiotemporal" approach, which has rarely been applied to the specific context of Dhaka's geography in a conference-level study.

3. Problem Statement & Objectives

3.1 Problem Statement

1. **Sparse Sensor Network:** Dhaka has very few reliable air quality monitoring stations (approx. 3-4 active sources), making it impossible to get accurate readings for every neighborhood (e.g., Mirpur vs. Uttara).

2. **Lack of Spatial Context:** Existing apps provide a city-wide average, which hides dangerous localized spikes in pollution.

3. **Static Predictions:** Current models predict static numbers, failing to visualize how pollution "clouds" travel across the city over time.

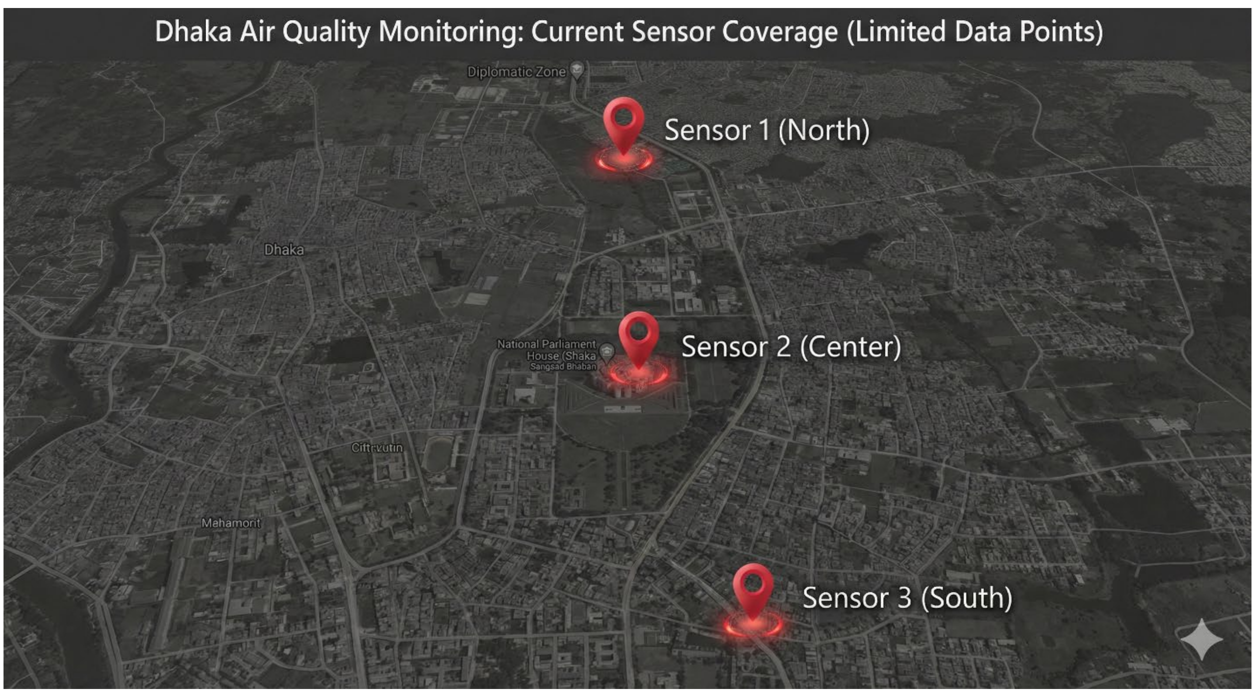

Figure 1: Current limitations of air quality monitoring in Dhaka, showing the scarcity of ground stations.

3.2 Objectives

The primary goal of this project is to build a predictive model that forecasts air quality maps.

1. **To Construct a "Virtual Sensor" Grid:** Implement Inverse Distance Weighting (IDW) to estimate pollution levels for unmonitored areas, creating a high-resolution 10x10 grid heatmap of Dhaka.
2. **To Develop a Spatiotemporal Model:** Train a Deep Learning model (Convolutional LSTM or similar architecture) that learns from past pollution maps to predict future pollution distribution.
3. **To Visualize Pollution Diffusion:** Create a visual representation (heatmap) that shows the predicted spread of PM2.5 pollutants for the next 24 hours, aiding in better decision-making.

4. Methodology

Our methodology follows a pipeline that converts raw point data into a visual prediction.

Step 1: Data Collection

We will collect historical PM2.5 and AQI data (2023–2025) from publicly available sources such as the US Department of State (AirNow) and CAMS. We will also integrate basic meteorological data (wind speed, temperature) if available.

Step 2: Preprocessing & Grid Generation (The "Virtual Sensor")

Since we only have 3-4 data points (stations) per timestamp, we cannot train a map-based model directly.

- We will define the geographical boundaries of Dhaka (Lat/Lon).
- We will divide the city into a grid (e.g., 10x10 or 20x20 blocks).
- Using **Inverse Distance Weighting (IDW)**, we will interpolate the values between the known stations to fill the grid. This converts our text data into "Image Data" (Heatmaps).

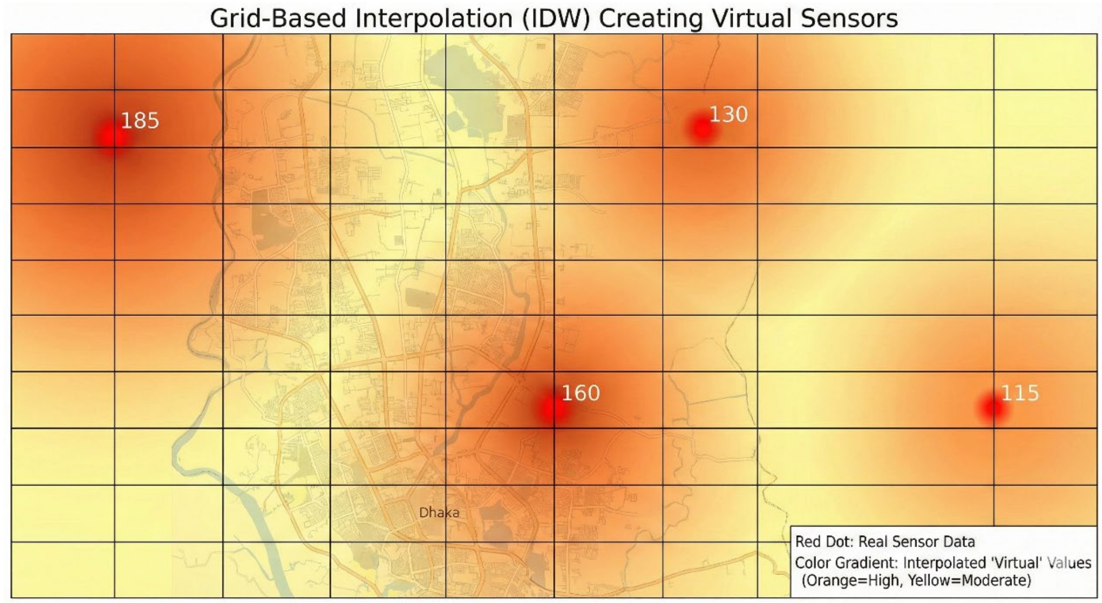

Figure 2: Visualization of the IDW Interpolation technique used to create 'Virtual Sensors' in unmonitored zones.

Step 3: Model Architecture (Spatiotemporal Learning)

We will use a Deep Learning architecture suited for video/sequence prediction, such as **ConvLSTM (Convolutional LSTM)**.

- **Input:** A sequence of the last 7 days' heatmaps
- **Process:** The Convolutional layers will extract spatial features (where the pollution is), while the LSTM layers will learn temporal patterns (how it changes over time).
- **Output:** A predicted heatmap for the future timestep.

Step 4: Evaluation & Visualization

The model will be evaluated using Root Mean Squared Error (RMSE) to compare the predicted pixel values against the actual interpolated values. The final output will be visualized using Python libraries (Matplotlib/Folium) as a color-coded map.

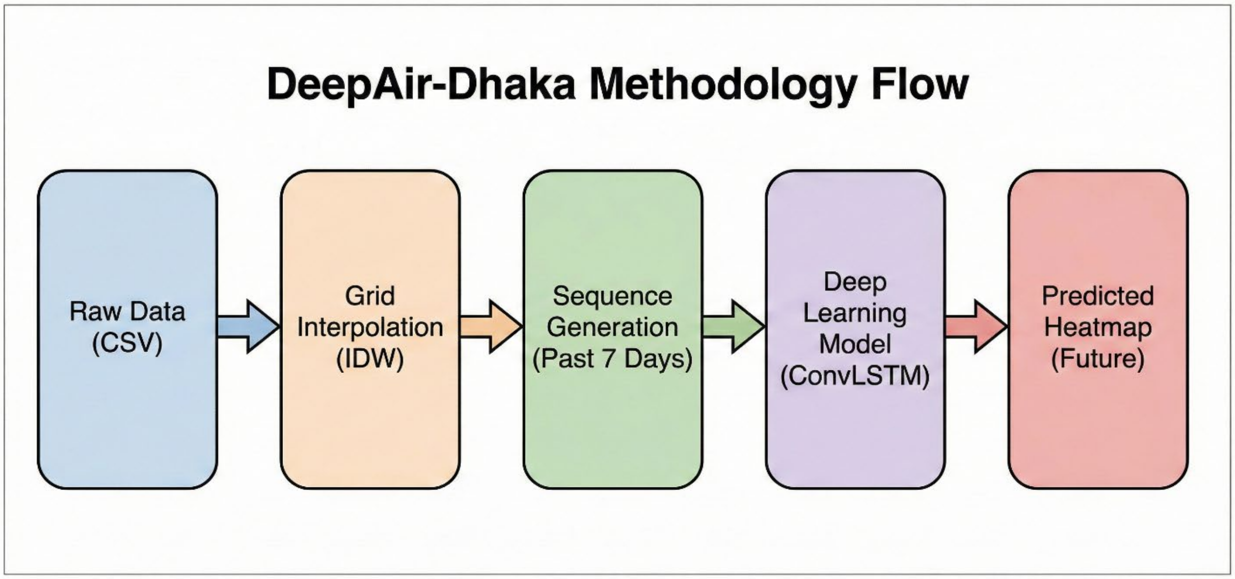

Figure 1: Data Processing and Modeling Pipeline.

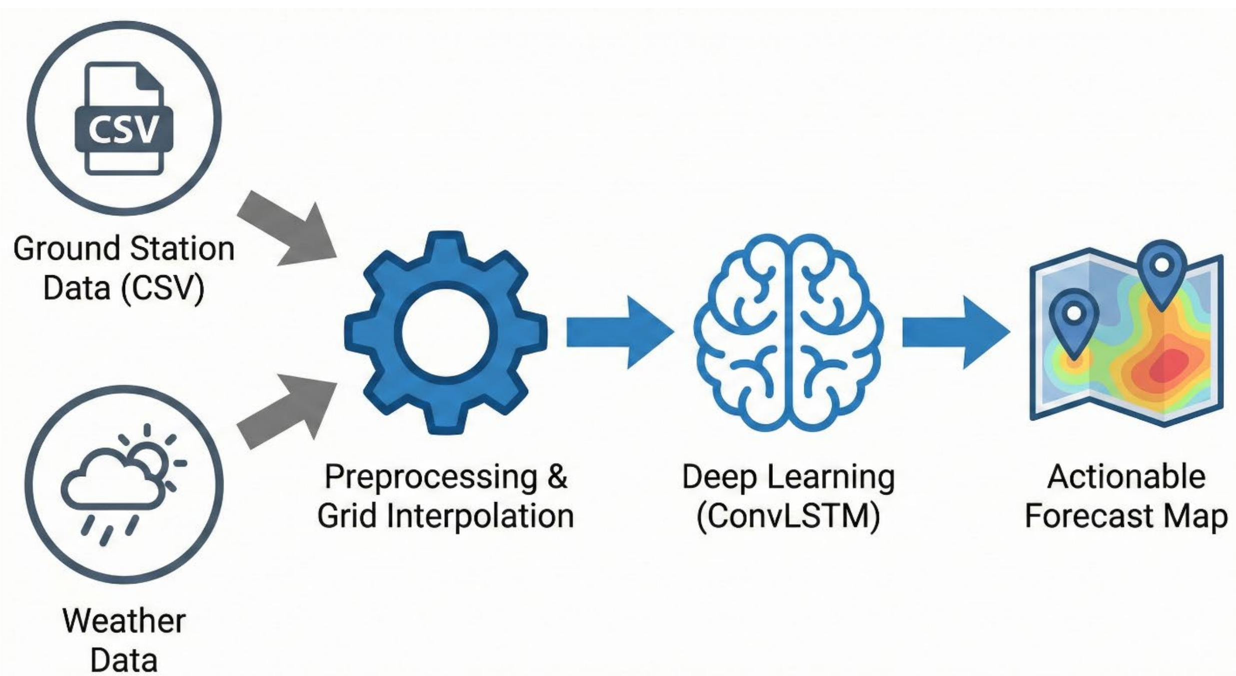

Figure 3: Complete System Pipeline of DeepAir-Dhaka, from raw data ingestion to spatiotemporal forecasting.

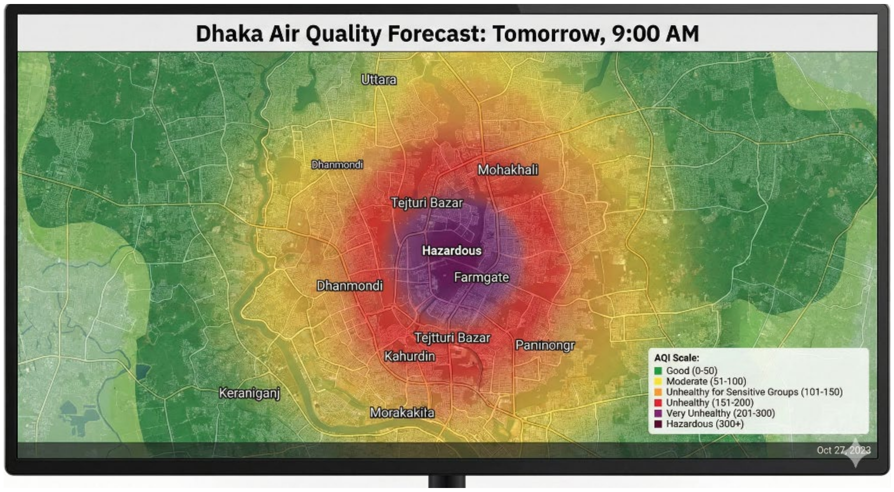

Figure 4: Simulated output of the DeepAir-Dhaka model, distinguishing high-pollution zones (Red/Purple) from green zones.

5. Conclusion

This project aims to bridge the gap between simple air quality monitoring and actionable urban insights. By shifting the focus from "City-Average" prediction to "Neighborhood-Level" mapping, DeepAir-Dhaka provides a novel approach to understanding pollution dynamics in a resource-constrained city.

Figure 5: Proposed web-based dashboard for public visualization of air quality forecasts.

Successfully implementing this model will not only demonstrate the power of Machine Learning in environmental science but also provide a foundation for future applications, such as identifying the specific sources of pollution spikes (e.g., industrial zones vs. traffic intersections). This research contributes to the growing field of AI for Social Good, offering a scalable solution for developing cities with limited sensor infrastructure.

5. References

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